ClipSAT : ACT Subject Test Math 1

Unit 0.01: ACT II Math Level 1 Diagnostic & Strategy Drill

Name: _____

Q1. In the standard x, y coordinate plane, the lines with equations 2x - y = 7 and x + y = 5 intersect at the point a, b. What is the value of a?

[ ] A: 1
[ ] B: 3
[ ] C: 4
[ ] D: 5

Q2. In right triangle ABC, altitude overline{BD} is drawn to hypotenuse overline{AC}. If AD = 4 and DC = 9, what is the length of altitude overline{BD}?

[ ] A: 6
[ ] B: 6.5
[ ] C: 13
[ ] D: 36

Q3. The bar graph below shows the number of students enrolled in each of four elective courses at a high school. If one student is selected at random from these four courses, what is the probability that the selected student is enrolled in Music?

[ ] A: dfrac{7}{50}
[ ] B: dfrac{7}{20}
[ ] C: dfrac{7}{13}
[ ] D: dfrac{13}{20}

Q4. The graph of the quadratic function fx = x^2 - 6x + 5 is shown in the standard x, y coordinate plane below. A horizontal line y = k intersects the graph of fx at two points that are 8 units apart. What is the value of k ?

[ ] A: 12
[ ] B: 8
[ ] C: 4
[ ] D: -3

Q5. In the standard x, y coordinate plane, the lines with equations 2x - y = 7 and x + y = 5 intersect at the point a, b. What is the value of a?

[ ] A: 1
[ ] B: 3
[ ] C: 4
[ ] D: 5

Q6. In the standard x, y coordinate plane, a circle has its center at -3, 5 and passes through the point 1, 8. What is the radius of this circle?

[ ] A: 5
[ ] B: 7
[ ] C: sqrt{13}
[ ] D: 25

Q7. If fx = 2x - 1 and gx = x^2 + 3 , what is the value of gf2 ?

[ ] A: 9
[ ] B: 12
[ ] C: 13
[ ] D: 21

Q8. What is the smallest positive integer n such that 126n is the square of an integer?

[ ] A: 7

[ ] B: 14

[ ] C: 21

[ ] D: 42